

**LAGOS STATE UNIVERSITY OF SCIENCE AND
TECHNOLOGY IKORODU**

COURSE: GENERAL WORKSHOP PRACTICE

COURSE CODE: LASUSTECH-GET 230

WORKSHOP MANUAL

ELECTRICAL ENGINEERING - OPTION

1. Title: Electrical Workshop Safety and Wiring Techniques

2. Objectives: At the end of the exercise, students should be able to:

- (i) Know various electrical hazards/safety measures
- (ii) Use Fire extinguisher effectively
- (iii) Know electrical rescue techniques
- (iv) Identify electrical tools, instruments and accessories
- (vi) Analysis simple electrical circuits.

3. Theory: Individual working in an electrical workshop or laboratory must be cautious of the hazards in the workplace and get to know the necessary measures at combating the effect of such occurrence. Knowledge of tools, instrument, accessories are fundamentals in the analysis of electrical system.

4. Equipment and Materials

- (i) Sample charts (ii) Fire extinguisher (iii) Sample of simple electrical circuits. (iv) Complete electrical tool box.

5. Procedures

(i) Presentation on electrical hazards, rescue techniques and fire extinguisher.

(ii) Presentation on tools and instruments- identification and application.

(ii) Presentation on Electrical Wiring Techniques- accessories, wiring methods and simple circuit analysis.

6. Questions.

- (a) List the three (3) steps to perform Cardio Pulmonary Resuscitation (CPR) (3 marks)
- (b)(i) Write the four (4) steps in using the fire extinguisher (2 marks)
- (ii) Write two (2) extinguishers suitable for combating electrical fire (2 marks)
- (c)(i) List two electrical wiring accessories (1 mark)
- (ii) Two (2) ways of maintaining electrical tools (1 mark)
- (ii) Briefly explain why lighting bulbs are connected in parallel (1 mark).

7. Conclusion

CHEMICAL ENGINEERING -OPTION

Title: Exploring the Fundamentals of Magnetic Stirrers: A Hands-on Introduction

Objectives:

1. To provide a practical introduction to the basic principles of magnetic stirrers
2. To familiarize students with the operation and application of magnetic stirrers in various fields
3. Understand the working principle of a magnetic stirrer.

Equipment/ Apparatus

1. Magnetic Stirrer
2. Beaker
- (3) Two fluids



Theory

Magnetic stirrers are widely used in various scientific disciplines, including chemistry, biology, and engineering, to mix and blend liquids efficiently. As a fundamental laboratory equipment, understanding the working principle and operation of magnetic stirrers is essential for students from diverse academic backgrounds. In this practical session, we will explore the basics of magnetic stirrers, and discuss its applications in different fields.

Magnetic stirrers are versatile equipment used in various laboratory and industrial settings, such as chemical synthesis, biological sample preparation, medical laboratory and materials processing.

A magnetic stirrer consists of a rotating magnet, a stirrer bar, and a container. The rotating magnet creates a magnetic field that induces rotation in the stirrer bar, which in turn mixes the liquid. The speed of rotation can be controlled to achieve the desired level of mixing.

Procedure

1. Assemble the magnetic stirrer kit according to the manufacturer's instructions.
2. Connect the power source to the rotating magnet.
3. Place the stirrer bar in the container and add water or a coloured liquid.
4. Observe and record the rotation of the stirrer bar as the power source is turned on.
5. Adjust the speed of rotation and observe the effects on the liquid.
6. Disassemble the setup and discuss the importance of magnetic stirrers in various applications.

Observation:

Precaution:

Conclusion:

Questions

1. What are the five (5) main applications of a magnetic stirrer? (5 marks)
2. Mention three (3) parts of a magnetic stirrer? (3 marks)
3. Explain the basic theory behind the working operation of a magnetic stirrer? (2 marks)

FOOD SCIENCE AND TECHNOLOGY OPTION

Title: Sieving Analysis

Aim: To perform an experiment on particle size analysis

Apparatus/Equipment: Milled corn powder, a set of sieves and shaker.

Theory: The various particles sizes in a flour for example can be separated and quantity in relation to the whole using a set of sieves (British, American and Tyler Sieve)

Procedure

Arrange the set of sieves in decreasing order of aperture size. Set up the arrangement on the shaker, put a known weight of the milled flour in the top sieve and activate the shaker for a reasonable time.

Observation:

Precaution

Conclusion

Exercises

- (i) What do you understand by sieving Analysis- 5 marks
- (ii) Mention the various mesh sizes that are used in the experiment- 3 marks
- (iii) Mention four (4) food samples that can be used to perform the experiment-(2marks).

AGRICULTURAL AND BIO- SYSTEMS ENGINEERING DEPARTMENT-OPTION

MODULE ONE

TITLE: WORKSHOP SAFETY PRECAUTIONS, SAFETY EQUIPMENT & PROPER CLOTHING

OBJECTIVES: TO BE ABLE TO KNOW THE SAFETY PRECAUTIONS, SAFETY EQUIPMENT & PROPER CLOTHING

MATERIALS: WELDING MACHINE, WELDING SHIELD, ELECTRODE, GAS CYLINDERS & ACCESSORIES, FIRE EXTINGUISHER & SAND BUCKET

PROCEDURE: STUDENT WERE THOUGHT PRECAUTIONS RELATING TO WELDING: ARC & GAS WELDING, PROCEDURE FOR THE USE OF FIRE EXTINGUISHER WAS HIGHLIGHTED AND SAFETY PRECAUTIONS WERE DISCUSSED

THEORY: WORKSHOP SAFETY PRECAUTIONS INVOLVES THE MAINTENANCE AND IMPROVEMENT OF STANDARD OF HEALTH SAFETY & WELFARE OF PEOPLE AT WORK.
TO BE ABLE TO CONTROL THE KEEPING AND THE USE OF DANGEROUS AND HIGHLY FLAMMABLE MATERIALS
STUDENTS ARE TO IMBIBED THE KNOWLEDGE OF PROPER CLOTHING

DIAGRAMS:

QUESTION: (1) LIST FIVE(5) WORKSHOP SAFETY PRECAUTIONS (5MARKS)

RESULTS:

OBSERVATION:

CONCLUSION:

MODULE TWO

TITLE: TRACTOR DRIVING (Daily Routine Checks)

OBJECTIVES: TO PUT TRACTOR (Vehicle) IN GOOD CONDITION BEFORE DRIVING

MATERIALS: TRACTOR

PROCEDURE: CHECKS WERE CARRIED OUT ON CRUCIAL PARTS OF TRACTOR'S PARTS CARRING
FLIUDS LIKE: FUEL, OIL, AIR & WATER

THEORY: FILLING OF FLUIDS TO ENGINE OIL THAT DROPS, BRAKE FLUID THAT DROPS,
WATER IN THE RADIATOR THAT DROPS AND AIR IN THE TYRE THAT REQUIRES
INFLATION IS QUITE NECCESARY AND IMPORTANT

DIAGRAMS:

QUESTIONS: (1) LIST FIVE (5) DAILY ROUTINE CHECKS ON TRACTOR (5MARKS)

RESULTS:

OBSERVATION:

CONCLUSION

MECHATRONICS ENGINEERING- OPTION

1.Title: Workshop Safety Precautions

2. Objectives: At the end of the exercise, students should be able to:

- (i) Know various safety precaution
- (ii) Identify safety kit
- (iii) Know how to use safety kits
- (vi) know when to raise alarm and converge at muster point

3. Theory:

4. Equipment and Materials: PPE and workshop safety tools

5. Procedures:

- (a) Presentation on safety precautions
- (b) Presentation on identification and use of safety kits
- (c) Presentation of the manners of responding to hazards

6. Results/observation

7. Questions-

- (a) Give eight (8) safety signs in workshop. (4marks)
- (b) Write four (4) safety kits in the workshop (2 marks)
- (c) Mention four (4) safety precautions while working on machine in the workshop. (4 marks)

8. Conclusion

CIVIL & CONSTRUCTION ENGINEERING -OPTION

1. Title: Block Moulding and Wood Work

2. Objectives: At the end of the exercise, students should be able to:

- Batch constituent materials for making block, i.e., cement, sand and water.
- Mix the proportional constituent materials to form homogenous sand mixture.
- Mould concrete block using appropriate mould.
- Identify different kind of wood
- Identify various kind of carpentry tools and their uses.

3. Theory: Sandcrete blocks are blocks made of cement, sand and water. They are used in house building and the construction of other structures such as drainage system, fence, manhole, etc. while Wood is a porous and fibrous structural tissue found in the stems and roots of trees and other woody plants.

4. **Equipment and Materials:** the tools and materials used include: Shovel, Head pan, Water can or bucket, Block mould (Steel), cement, sand, water, Marking Knife, Straight Edge, Tri Square, Pencil, Tape Rule, Chisel, Plane, Saw, Scraper, Hammer, Nail punch, Mallet, Pincher, Clamp, Bench Hook etc.

5. **Procedures:** the procedures involved in block making include batching, mixing, moulding, curing, stacking and transporting, while wood work involves identification of simple carpentry tools.

6. **Results:** records of block making should be made, sketching carpentry tools and stating their uses.

7. Questions:

- i. Briefly state the stages and procedures for block making . 5mks

- ii. Sketch five (5) simple tools used in wood works and state their uses. 5mks

8. **Conclusion:** There are other forms of blocks in terms of materials and making processes, but sandcrete blocks are the most used in construction. Both block moulding and wood work are lucrative and high in demand, therefore should be done bearing in mind their qualities.

MECHANICAL ENGINEERING - OPTION

Module 1

1. Title: Safety Precautions in Mechanical Engineering workshops

2. Objectives: At the end of the exercise, students should be able to:

- (a) Know various mechanical hazards/safety measures
- (b) Know safety precautions as applicable to:
 - (i) Personal safety
 - (ii) Emphasize tidiness in workplaces
 - (ii) Fitting and benchworks
 - (iii) Welding shop activities
 - (iv) Automobile shop and activities
 - (v) Refrigeration and A/C uses
- (c) Use Fire extinguisher effectively
- (d) Identify fitting tools, instruments and accessories

3. Theory:

4. Equipment and Materials

5. Procedures

6. Results

7. Questions

- 1. What is the first approach to safety in work place?
- 2. Mention the usefulness of the following PPEs
 - (a) Nose mask
 - (b) Ear muff
 - (c) Safety booth
 - (d) Goggle

8. Conclusion

Module 2

Title: Automobile services and maintenance activities

Objectives: At the end of the exercise, students should know;

1. Layout and structure of a vehicle consisting of the followings:
 - i Engine
 - ii Clutch
 - iii Transmission
 - iv Propeller shaft
 - v Differential
 - vi Rear axle
2. Function of each of the sub-systems listed above and draw the layout of a motor vehicle
3. Daily maintenance needs/checks on a vehicle before moving on the road;
 - i Water/coolant in the radiator
 - ii Gauging of engine oil
 - iii Checking/gauging the pressure of tires
 - iv Ensure availability of all safety needs such as extra tire, wheel spanner, jack, fire - extinguisher, C- caution triangles, gallon of water, some useful spanners, etc.
4. Learn and demonstrate replacement of flat tires.
5. Understand and carry out necessary checks on road tires.
6. Understand the purpose of wheel balancing and alignment.

Question.

1. What are the daily routine checks to be made on a vehicle before embarking on a journey?
2. List the steps involved in changing the flat tire of a car

Module 3

1. Understand operational principles of Domestic Refrigerator and Air conditioner (R and A) systems
2. Know the major components of R and A systems and their respective functions.
3. Know the safety handling methods and precautions in the use of R and A systems.

Question

1. List any three of the major components of a domestic refrigerator.

COMPUTER ENGINEERING – OPTION

Cyber Security Practical Course Manual

Aim: to provide foundational knowledge of cybersecurity, practical threat mitigation skills, awareness of digital protection, and ethical practices.

Objective: To be able to understand cybersecurity concepts, recognizing cyber-attacks, implementing basic security measures, and promoting safe online practices.

Module 1: Understanding Cyber Security Fundamentals

CIA Triad involves three core principles: Confidentiality, which restricts access to information; Integrity, which ensures data accuracy; and Availability, which ensures resource access. Risk Management encompasses identifying, assessing, and mitigating risks such as threats and vulnerabilities. Threats are potential causes of harm, while vulnerabilities are weaknesses that can be exploited. Phishing involves deceptive attempts to obtain sensitive information, malware refers to malicious software that disrupts or gains unauthorized access, and DoS attacks aim to make resources unavailable.

The procedure for ensuring confidentiality includes encrypting text files and accessing them without keys. Integrity is maintained by creating and comparing file hashes, while availability is tested by simulating DoS attacks. Risk management involves identifying and assessing risks and developing mitigation plans. In conclusion, understanding the CIA Triad and risk management is crucial for effective cybersecurity measures. The learning outcomes include applying encryption for confidentiality, hashing for integrity, understanding DoS attacks' impact on availability, and gaining insights into risk mitigation.

Module 2: Simulating a Cyber Attack

Objective: The objective is to simulate and understand phishing and Man-in-the-Middle (MitM) attacks.

Theory: Phishing involves deceptive attempts to obtain sensitive information, while MitM attacks intercept and possibly alter communication between parties. Types of MitM attacks include eavesdropping, session hijacking, packet injection, and SSL stripping.

Procedure: Phishing simulation involves creating and sending mock phishing emails. MitM simulation entails setting up a Wi-Fi network to intercept and monitor traffic.

Conclusion: These simulations highlight the importance of user education and secure communication channels.

Learning Outcome: The learning outcomes include recognizing phishing signs, understanding MitM operations, and appreciating the importance of secure communication.

Module 3: Protection against Malware

Objective: The objective is to explore and implement measures for protecting against malware.

Theory: Malware is software designed to disrupt or gain unauthorized access. Antivirus software detects and removes malware. Best practices for protection include updating software, using strong passwords, practicing cautious email and download habits, ensuring browser security, and performing regular backups.

Procedure: Using antivirus involves installing antivirus software, running malware samples, and observing detection and removal. Implementing secure practices includes updating software, setting up firewalls, practicing safe browsing, creating strong passwords, and enabling multi-factor authentication (MFA).

Conclusion: Combining antivirus software with secure practices is essential for effective malware protection.

Learning Outcome: Enhanced security is achieved through regular updates, cautious online behavior, and strong authentication mechanisms.

Module 4: Cryptography

Introduction to Cryptography: Basic concepts include encryption, which converts plaintext to ciphertext; decryption, which reverses the process; and hashing, which ensures data integrity. Encryption protects data confidentiality, decryption allows authorized access to the original data, and hashing produces unique hash values for unique input data.

Practical Cryptography: Encrypting and decrypting data involves choosing an algorithm (AES for symmetric encryption, RSA for asymmetric encryption), generating keys (shared secret or public-private pair), encrypting data, transmitting it securely, and decrypting it at the recipient's end. For example, Alice sends a confidential file to Bob using AES encryption.

Learning Outcome: The learning outcome is the ability to apply cryptographic principles to enhance security and protect sensitive information.

Exercises:

1. What are the steps involved in risk management, and how do you develop a risk mitigation plan to address identified threats and vulnerabilities?
2. What are the different types of Man-in-the-Middle (MitM) attacks, and how does each type exploit communication vulnerabilities?
3. What are the best practices for protecting against malware, and how do antivirus programs contribute to these practices? Provide examples of effective strategies.
4. Explain the difference between encryption, decryption, and hashing. How does each process contribute to data security?
5. In what scenarios would you use AES versus RSA encryption, and how do the key management processes differ between symmetric and asymmetric encryption?